

Mobile Development

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Code 1: GridLayout

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:columnCount="3"
    android:rowCount="3"
    android:padding="16dp"
    android:alignmentMode="alignMargins"
    android:layout_marginTop="50dp">

    <Button
        android:id="@+id/btn1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_column="0"
        android:layout_row="0"
        android:layout_columnWeight="1"
        android:text="1" />

    <Button
        android:id="@+id/btn2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_row="0"
        android:layout_columnWeight="1"
        android:text="2" />

    <Button
        android:id="@+id/btn3"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_row="0"
        android:layout_columnWeight="1"
        android:text="3" />

    <Button
        android:id="@+id/btn4"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_column="0"
        android:layout_row="1"
        android:layout_columnWeight="1"
        android:text="4" />
```

```
<Button
    android:id="@+id/btn5"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_row="1"
    android:layout_columnWeight="1"
    android:text="5" />
```

```
<Button
    android:id="@+id/btn6"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="2"
    android:layout_row="1"
    android:layout_columnWeight="1"
    android:text="6" />
```

```
<Button
    android:id="@+id/btn7"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="0"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="7" />
```

```
<Button
    android:id="@+id/btn8"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="8" />
```

```
<Button
    android:id="@+id/btn9"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="2"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="9" />
```

```
</GridLayout>
```

```
package com.example.project1java;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
```

```

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        // Find buttons and set click listeners
        for (int i = 1; i <= 9; i++) {

            int resId = getResources().getIdentifier(
                "btn" + i,
                "id",
                getPackageName()
            );

            Button button = findViewById(resId);

            final int number = i;

            button.setOnClickListener(new View.OnClickListener() {
                @Override
                public void onClick(View v) {

                    Toast.makeText(
                        MainActivity.this,
                        "Button " + number + " clicked",
                        Toast.LENGTH_SHORT
                    ).show();

                }
            });
        }
    }
}

```

Code 2:FrameLayout

```

<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/imgBackground"
        android:scaleType="centerCrop"
        android:src="@drawable/pic"/>

```

```

        <TextView
            android:id="@+id/tvTitle"
            android:layout_width="177dp"
            android:layout_height="79dp"
            android:background="#800000"
            android:gravity="center"
            android:padding="12dp"
            android:text="Welcome to android"
            android:textColor="@android:color/black"
            android:textSize="24sp"
            android:textStyle="bold" />

    </FrameLayout>

```

```

package com.example.project1java;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        // Find buttons and set click listeners

    }
}

```

Code 3: ConstraintLayout

```

<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Heading -->
    <TextView

```

```

        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to\nConstraintLayout"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginTop="32dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

<!-- Image -->

<!-- Name -->
<ImageView
    android:id="@+id/imageView"
    android:layout_width="405dp"
    android:layout_height="151dp"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="16dp"
    android:scaleType="centerCrop"
    android:src="@drawable/pic"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/tvTitle" />

<EditText
    android:id="@+id/etName"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:hint="Enter name"
    android:padding="12dp"
    android:inputType="textPersonName"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:layout_marginTop="25dp"
    app:layout_constraintTop_toBottomOf="@id/imageView"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<!-- Email -->
<EditText
    android:id="@+id/etEmail"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:hint="Enter email"
    android:padding="12dp"
    android:inputType="textEmailAddress"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:layout_marginTop="15dp"
    app:layout_constraintTop_toBottomOf="@id/etName"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

```

```

<!-- Submit -->
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:text="SUBMIT"
    android:textSize="14sp"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:layout_marginTop="15dp"
    app:layout_constraintTop_toBottomOf="@id/etEmail"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<!-- Login -->
<TextView
    android:id="@+id/tvLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Already have an account? Login"
    android:textSize="12sp"
    android:textColor="#303F9F"
    android:layout_marginTop="12dp"
    app:layout_constraintTop_toBottomOf="@id/btnSubmit"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

```

package com.example.project1java;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etName, etEmail;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(v -> {

```

```
String name = etName.getText().toString();
String email = etEmail.getText().toString();

Toast.makeText(
    MainActivity.this,
    "Name: " + name + "\nEmail: " + email,
    Toast.LENGTH_SHORT
).show();
});
}
```